RIPHAH INTERNATIONAL UNIVERSITY

LAIBA MUGHAL

70077

BS SE – 3

Data Structures Lab

Lab No# 14

Task 1: Tree Traversals

Given Tree:

```
      A
    /  \
   B    C
  / \    \
 D   E    F
```

Root Node :

The root node is the topmost node of the tree.

So, the root node is A.

Leaf Nodes:

Leaf nodes are the nodes that do not have any children.

So, the leaf nodes are D, E, and F.

Degree of B:

The degree of a node is the number of its children.

Node B has two children (D and E).

So, the degree of B is 2.

Height of Tree:

Height is the number of edges in the longest path from root to a leaf node.

Longest path: A → B → D (or A → B → E or A → C → F)

This path has 2 edges.

So, the height of the tree is 2.

Level of Node E :

Level starts from 0 at the root.

A → Level 0

B, C → Level 1

D, E, F → Level 2

So, the level of node E is 2.

Task 2: Tree Traversals

Given Tree:

```
    6
   /  \
  3    10
 / \
2   5
```

1. In order Traversal (Left, Node, Right)

Step:

Left of 6 → subtree of 3 → (2, 3, 5)

Then 6

Then right → 10

So, In order = 2, 3, 5, 6, 10

2. Preorder Traversal (Node, Left, Right)

Step:

Start from 6

Then left subtree → 3, 2, 5

Then right → 10

So, Preorder = 6, 3, 2, 5, 10

3. Post order Traversal (Left, Right, Node)

Step:

Left subtree → 2, 5, 3

Right subtree → 10

Then root → 6

So, Post order = 2, 5, 3, 10, 6

Task 3: Insert into BST

Insert values: 45, 25, 65, 10, 30

Step 1: Insert 45

45

Step 2: Insert 25

25 is less than 45, so it goes to the left.

```
  45
 /
25
```

Step 3: Insert 65

65 is greater than 45, so it goes to the right.

```
  45
 /  \
25   65
```

Step 4: Insert 10

10 is less than 45 → go left

10 is less than 25 → go left

```
   45
  /  \
 25   65
 /
10
```

Step 5: Insert 30

30 is less than 45 → go left

30 is greater than 25 → go right

```
    45
   /  \
  25   65
  / \
10  30
```